



Cours Réseaux : SLAAC/DHCPv6

IPV6

● Introduction

● Principes

- NDP-RA
- DHCPv6

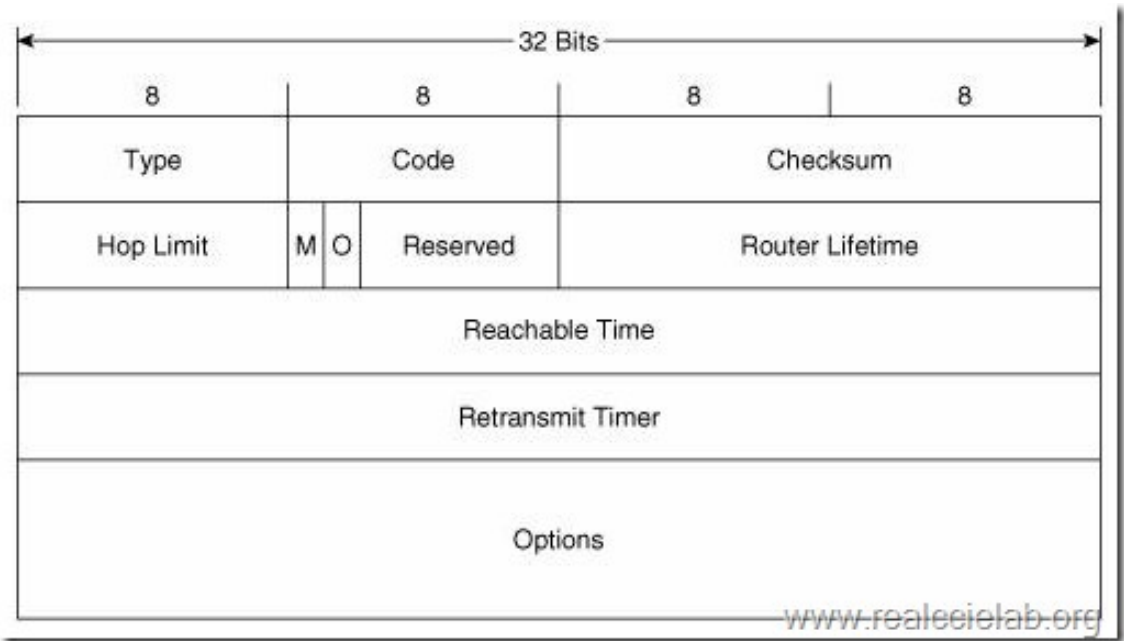
● Configuration IOS de base

● TP

IPv6 : Adressage Dynamique

NDP

- RA annonces du routeur
 - Par défaut SLAAC
 - Bit M (Manage) = DHCPv6
 - Bit O (Other) = Stateless DHCP



Prefix Information Option

Prefix Information Option (Type 3)

Fields and Meaning

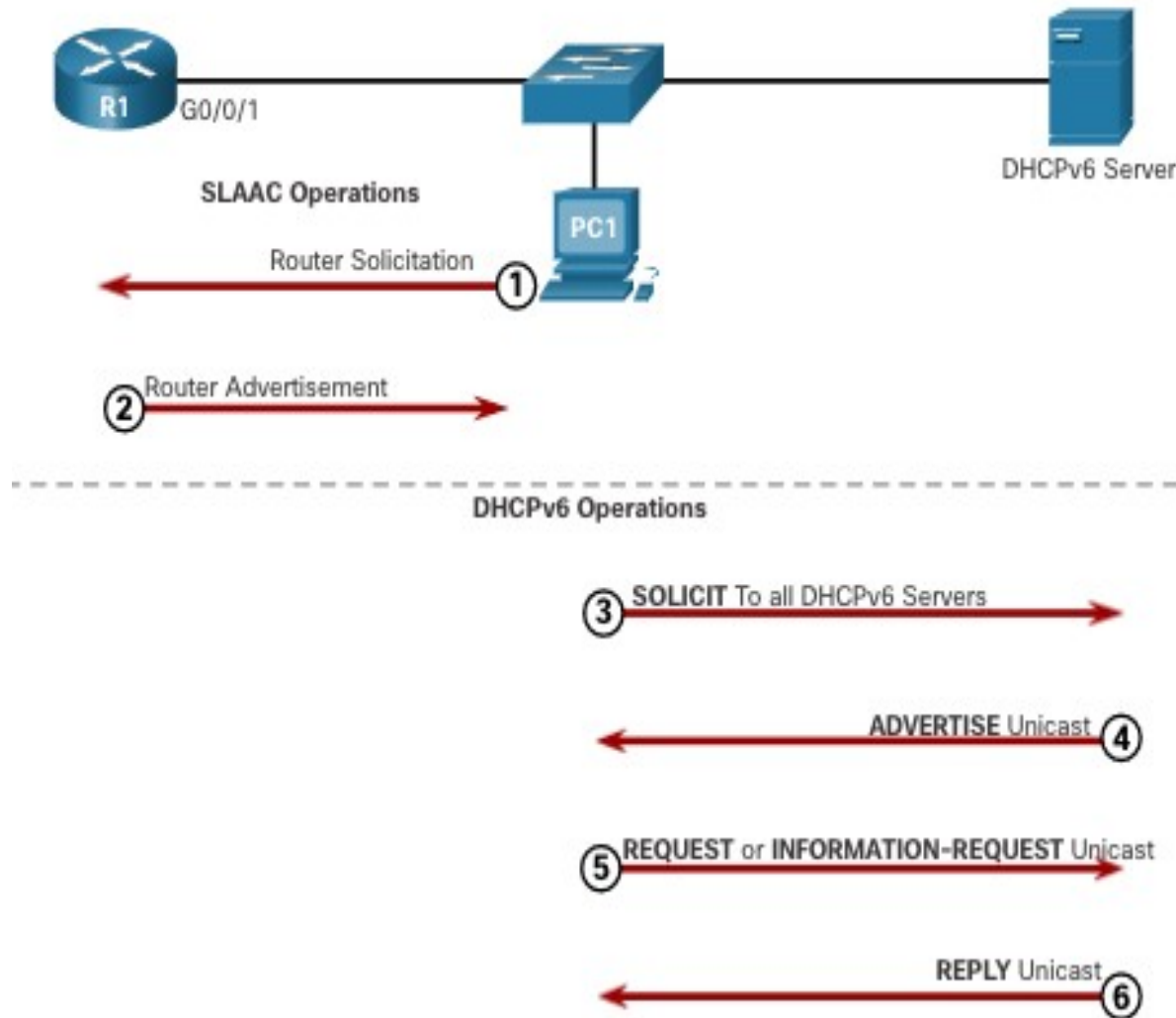
Field	Size	Meaning
Type	8 bits	Value = 3 (Prefix Information)
Length	8 bits	Length of the option (always 4, i.e. 32 bytes)
Prefix Length for /64)	8 bits	Length of the IPv6 prefix (e.g., 64
L (On-link flag) on-link (local network)	1 bit	1: the prefix is considered
A (Autonomous flag) prefix can be used for autonomous address configuration (SLAAC)	1 bit	1: the
Reserved	6 bits	Reserved (set to 0)
Valid Lifetime valid	32 bits	Time during which the prefix is
Preferred Lifetime which the prefix is preferred for creating new addresses	32 bits	Time during
Reserved2	32 bits	Reserved (set to 0)



Ipv6 avec PIO

- A=1 SLAAC
 - NDP
- O=1 Stateless DHCP
 - NDP
 - Autres infos (DNS) avec DHCPv6
- M=1 Statefull DHCP
 - Tout en DHCPv6
- A=1 et M=1
SLAAC et DHCPv6

Mode opératoire en DHCPv6



Configuration Statefull DHCP

```
R1(config)# int g0/0/1
R1(config-if)# ipv6 nd managed-config-flag
R1(config-if)# ipv6 nd prefix default no-autoconfig
R1(config-if)# end
R1#
R1# show ipv6 interface g0/0/1 | begin ND
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use DHCP to obtain routable addresses.
R1#
```




```
R1(config)# ipv6 dhcp pool IPV6-STATEFUL
R1(config-dhcpv6)#
```

```
R1(config-dhcpv6)# address prefix 2001:db8:acad:1::/64
R1(config-dhcpv6)# dns-server 2001:4860:4860::8888
R1(config-dhcpv6)# domain-name example.com
R1(config-dhcpv6)#
```

```
R1(config-if)# ipv6 dhcp server IPV6-STATEFUL
```

Configuration Stateless

```
R1(config-if)# ipv6 nd other-config-flag
R1(config-if)# end
R1#
R1# show ipv6 interface g0/0/1 | begin ND
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
  Hosts use DHCP to obtain other configuration.
R1#
```



```
R1(config)# ipv6 dhcp pool IPV6-STATELESS
R1(config-dhcpv6)#
```

```
R1(config-dhcpv6)# dns-server 2001:db8:acad:1::254
R1(config-dhcpv6)# domain-name example.com
R1(config-dhcpv6)# exit
R1(config)#
```

```
R1(config-if)# ipv6 dhcp server IPV6-STATELESS
R1(config-if)# no shut
```



```
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 dhcp relay destination 2001:db8:acad:1::2 G0/0/0
R1(config-if)# exit
R1(config)#
```

Conclusion

- Concepts de Base
 - SLAAC/NDP
 - Stateless DHCP
 - Statefull DHCP
 - Relais DHCP
- Configuration en IOS
- TP